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Does variation in garden characteristics influence the conservation of birds in suburbia?

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A B S T R A C T

Can enhancement of garden habitat for native birds have conservation benefits, or are garden bird assemblages determined by landscape and environmental characteristics? The relative roles of vegetation structure, floristics and other garden attributes, and environmental and landscape controls, on the abundance and richness of bird species in 214 back or front gardens in 10 suburbs of Hobart, Tasmania, Australia, are addressed to answer this question. Birds were counted in each garden and the resources they utilized noted. Vascular plant species and other attributes of the garden were noted, along with rainfall, altitude, distance from natural vegetation, distance from the city and garden size. Garden floristics and bird assemblages were ordinated, and garden groups characterized by particular assemblages of birds identified. General linear modelling was used to determine the combinations of independent variables that best predicted the richness of birds and the abundance of individual bird species and groups of species. The models for bird richness, bird species and groups of bird species were highly individualistic. Although native birds showed a preference for native plants, they also utilized many exotic plants. Exotic birds largely utilized exotic plants. Variation in garden characteristics does substantially affect the nature of garden bird assemblages in Hobart, with weaker environmental and landscape influences. The fact that gardens can be designed and managed to favour particular species and species assemblages gives gardeners a potentially substantial role in the conservation of urban native avifauna.

1. Introduction

There are a growing number of studies on assemblages of birds in cities. This work has indicated that: urban bird communities are characterised by less species richness and diversity than communities in adjacent natural vegetation; more biomass and density; and, the super-dominance of a few species, which are usually exotics (Jones, 1981; Beissinger and Osborne, 1982; Green, 1986; Catterall et al., 1989; Clergeau et al., 1998). These tendencies vary within urban areas along the gradient from natural vegetation to heavily urbanised areas (Green, 1986; Jokiamaki and Suhonen, 1993; Germaine et al., 1998; Fernandez-Juricic, 2000; Chamberlain et al., 2004; however, see Catterall et al., 1989 for an exception). Bird species richness in gardens is affected by tree species composition, with deciduous rather than coniferous trees promoting the highest bird species richness in Europe (Thompson et al., 1993). Garden area has a positive influence on garden bird species richness (Thompson et al., 1993; Chamberlain et al., 2004). Clumped trees result in more birds per unit area than dispersed trees (Day, 1995).

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In recent years, many Australian books have recommended native plants, which are believed to attract native birds, for gardens (Dengate, 2000; Pizzey, 2000; Grant, 2003; Burke, 2004). However, very little quantitative data has been gathered in Australia, or elsewhere, to determine whether native plants, or indeed any type of plant, do, in fact, increase the number and richness of native birds utilising a garden. Day (1995) found that, although native plants were more attractive to a wider range of bird species than introduced plants in Hamilton, New Zealand, they did not necessarily attract a greater abundance of birds, and noted that a number of introduced bird species seemed to be able to meet their habitat requirements equally well from native or introduced plants. In Australia, Catterall et al. (1989) found that neither silvereyes (*Zosterops lateralis*) nor exotic species displayed any clear foraging preferences for groups of plant species defined by origin or physiognomy. On the other hand, there were stronger associations of bird use with plant species and genera. French et al. (2005) found that the Australian native plant genera *Banksia* and *Grevillea* were significantly more attractive than the exotic genera *Hibiscus* and *Camellia* to both categories of all birds and Australian native species.

Very little work has been done to quantify the effects of non-vegetation garden components, such as pets, bird baths, and supplementary feeding, on garden bird assemblages. Lepczyk et al. (2004) investigated the number of people taking part in behaviours that could affect bird abundances, but did not quantify the effects of any of these behaviours. Gillies and Clout (2003) found that domestic house cats *Felis catus* regularly prey upon birds in Auckland, New Zealand, but there has been no attempt to quantify the effects of cat presence on garden bird abundance. Thompson et al. (1993) found that feeding frequency had no major effect on the number of bird species recorded in a garden, whereas Chamberlain et al. (2004) found that the probability of occurrence of many species increased in the presence of supplementary feeding.

Clergeau et al. (2001), on examination of urban bird data from France, Finland and Canada, found that, in winter at least, the composition of urban bird communities differed from that of surrounding 'periurban' landscapes; implying that local features were more important than landscapes. However, Chamberlain et al. (2004) found few associations between birds and garden habitat, suggesting that surrounding habitat is one of the main determinants of occurrence in gardens for most bird species, although the surrounding habitat analysed by Chamberlain et al. (2004) was within 100 m of the gardens and thus could still be considered local vegetation, rather than landscape. Thus, there is still uncertainty on whether enhancement of garden habitat is worthwhile, or if urban bird assemblages are simply determined by landscape and environmental features.

The main aim of the present paper is to determine if garden variation does affect garden bird composition, abundance and richness. If garden birds are primarily influenced by surrounding landscapes, it is unlikely that garden level habitat enhancement can significantly increase urban bird diversity. Conversely, if urban birds are primarily influenced by garden features and are, to some extent, independent of the surrounding landscape, domestic gardens could become very important in the conservation of urban avifauna. If variation in gardens accounts for at least some of the variation in garden birds, it is important to understand the nature of this relationship. Therefore, this paper relates attributes of garden birds to attributes of gardens to determine which garden attributes best explain bird species richness, bird assemblage abundances, bird group abundances and individual species abundances. Particular focus is given to the hypothesis of a strong positive relationship between local native plants in gardens and the occurrence of native bird species.

2. Methods

2.1. Study area and garden selection

Ten suburbs of Hobart, Tasmania, Australia, were selected for sampling (Fig. 1). These represented a range of environments (annual precipitation and altitude) and locations (distance to Hobart and distance to bush). The suburbs range between: 5 and 370 m above sea level; 1 km and 16 km (as the bird flies) from the Hobart General Post Office; <5 m to greater than 3 km from natural vegetation; and receive between 500 and 1000 mm of mean annual precipitation. Hobart is a linear city of 190,000 people, situated on both banks of a large estuary (Fig. 1), and is contained to a large degree by forested hills and mountains. Most people live in self-contained houses, with both front and back gardens. These gardens vary markedly in their structural characteristics and floristic composition, both within and between suburbs (Daniels and Kirkpatrick, in press; Kirkpatrick et al., in press).

Within the suburbs, the number and location of gardens used in the study depended on the willingness of owners to allow observation. Almost all owners (>90%) were happy to have the survey done on their properties. Within suburbs, gardens were selected subjectively without preconceived bias (Mueller-Dombois and Ellenberg, 1974). The total number of properties from which data were collected was 107. With the front and back gardens being treated as independent locations, the total number of sites was 214. Treating the front and back gardens of a property as independent data points means that it is likely that the same individual birds were often recorded in both gardens. However, multiple recordings of individuals birds were not considered a problem because, no matter how close survey points were located, it was hypothesised that birds would only visit a garden (and thus be recorded at that data point) if there was suitable habitat present. This was further ensured by only recording species that stopped and utilised the garden in some manner (see below). The gardens range between 50 and 1600 m^2 in area.

2.2. Data collection

Stationary point counts have special value in studies of bird–habitat associations, when habitat variables are measured at the counting points (Bibby et al., 2000), and there is insufficient space for transects, as in the present study. Twenty minute stationary point counts were undertaken between sunrise and 1100 hours, 1100 hours and 1400 hours, or 1400 hours and sunset. A few pilot surveys identified 20 min as a sufficiently long period to capture all individuals utilizing a garden on any one occasion. Nonetheless, it is unlikely that sampling

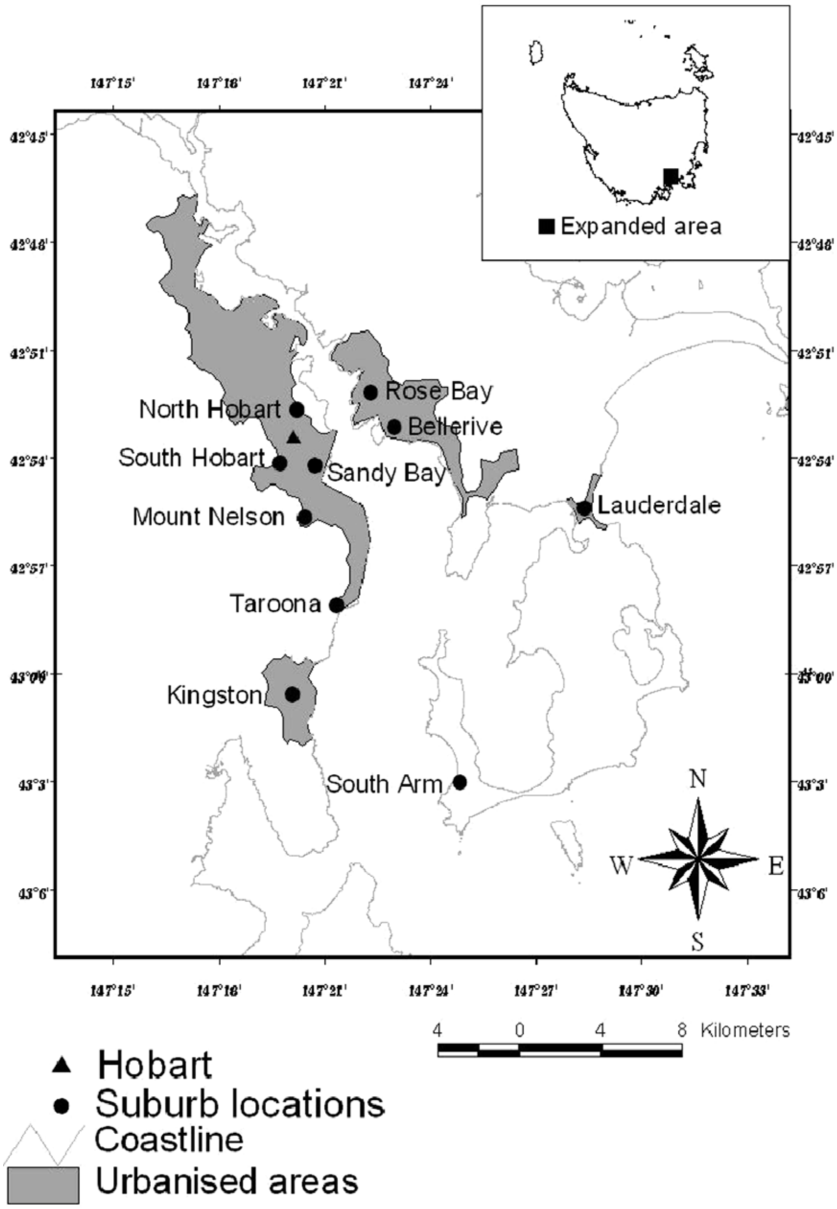

Fig. 1 – Locations of the suburbs used for sampling.

captured every individual or species utilising a garden, but as the focus of the project was not to establish population estimates, but simply to compare the relative attractiveness of different garden types, it was considered that spending an equal amount of time in each of the gardens was more important than the amount of time spent. Each garden was sampled 4 times throughout the course of the project, twice in winter 2004 and twice in spring 2004. Each garden was sampled no more than twice in any one of the time periods, and was sampled in each of the time periods. Gardens were never sampled twice in one day. Surveys were not carried out in rain, or when wind was strong enough to be a potential impediment to accurate aural identification. A location that maximised visibility and minimised the conspicuousness of the observer was chosen, and used for all observations.

Species utilising the garden were identified by sight and/or call, and their abundance, and use of plant species recorded in the classes: foraging on nectar; foliage gleaning; bark gleaning; foraging on fruit; nesting. If an individual utilised multiple plant species, and/or utilised a plant species in a number of ways, all were recorded. Individuals that flew through or over gardens and did not stop and use the garden in some way were ignored. For birds that primarily forage by catching invertebrates on the wing, such as tree martins (*Hirundo nigricans*) and welcome swallows (*Hirundo neoxena*), it can be very difficult to distinguish between the acts of simple flight and swooping in the search of invertebrates. Thus, for these species an arbitrary measure was decided upon: all individuals recorded above tree level (approximately 30 m) were considered to be 'flying', and thus ignored, and all individuals tree level and below were deemed to be foraging. Bird species nomenclature follows Christidis and Boles (1994).

A list was made of all observable vascular plant species in each of the front and back yards of the gardens. Plants that were deemed inaccessible to birds, such as those enclosed in conservatories, greenhouses or encased in mesh, were not included in the species list. Plants that were based in adjacent yards but were growing into, or overhanging the garden being surveyed were included. Non-indigenous conifers under the height of 4 m were grouped into a broad class, dwarf conifer species. Due to their ubiquity, voluntary herbaceous weeds were excluded from the lists, whilst voluntary shrubs and trees were included. Due to difficulties with discrimination when tightly mown, species that only occur in the mosaic of species that comprise suburban lawns were excluded. A point that was considered to be roughly halfway between the front and back garden was chosen, and plants either side of this point were included in the respective compositions of the yards. Plants that were invisible from the bird observation point were included.

The height of the tallest stratum (regardless of percentage cover) was estimated into categories: 0–2 m; 2–4 m; 4–8 m; 8–30 m. The percentage of total site covered by: evergreen trees (all evergreen vegetation >4 m in height); deciduous trees (deciduous vegetation >4 m in height); large shrubs (woody plants between 2 and 4 m in height); small shrubs (woody plants less than 2 m in height); lawn; artificial surfaces (concrete, gravel, small buildings, swimming pools etc.) and herbaceous vegetation, were estimated into the categories: 0–10%, 10–25%, 25–50%, 50–75%, 75–100%.

At the time of each bird survey, the number of cats and dogs present in the yard were recorded, as were the presence of chicken yards, water supplies and supplementary food. At the completion of the four bird surveys, the records for chicken yards, water supply, supplementary food and dogs were expressed in qualitative form. For permanent water supply, it was deemed that water should be available on 3 or more of the bird survey periods – this was chosen so as to avoid bias from ephemeral water supplies such as flooded gutters or puddles. Swimming pools were not considered as water supplies. For supplementary feeding, all that was required was for some kind of food to be intentionally supplied at the time of at least one of the bird surveys. Dogs were converted to presence/absence data due to the fact that the number of sites with more than one dog was low. Cat data were expressed as a cumulative total of cats recorded throughout the four surveys, as there was substantial variation in numbers of cats visiting from adjacent properties. All bird abundance data are the sum of the number of individuals counted in the four sampling periods.

2.3. Data analysis

Data were collected separately from front and back gardens on the same blocks, because these typically exhibit very different characteristics (Daniels and Kirkpatrick, in press). If garden characteristics are important in determining bird species composition, the proximity of gardens should have little or no effect. This proposition was tested by the following process. The Bray–Curtis dissimilarity values for bird species abundance for all pairs of gardens were calculated. All structural and environmental variables, plant species richness and the four dimension ordination scores for plant species composition were standardised between 0 and 1, and these standardised values used to calculate Bray–Curtis dissimilarity values for garden characteristics between all pairs of gardens. Multistate class variables were disaggregated into qualitative variables for this analysis. The dissimilarity values for bird species composition were subtracted from the dissimilarity values for garden characteristics. The product of this subtraction was compared between front and back gardens on the same property and all other garden pairs, using oneway ANOVA. If proximity was a major influence on similarity in bird species composition and, therefore, the samples were not independent, a significant difference between these two sets of values would be expected, with the value being less for the gardens on the same property.

Abundance data were used to ordinate the bird species compositions of the garden sites using global non-metric multidimensional scaling, following the default options in DECODA (Minchin, 2001). This technique has been shown to produce the least distortions with data with large numbers of zeros (Minchin, 1987). The same process was used with presence/absence data for vascular plant taxa. The four dimensional ordination scores for birds (stress = 0.065448 (6.5448%)) were used as inputs for an agglomerative classification using Ward's method and Euclidean distance in Minitab (Minitab Inc., 1998). The classification of the bird assemblages was accepted at a level of five groups (similarity level = 39.15%), on the basis of a break in the slope of similarity levels with successive fusions.

Pearson's product moment correlation coefficient was used to test the strength of linear relationships between continuous variables. One way ANOVA was used to determine relationships between class and continuous variables. A level of $P < 0.05$ was taken to denote a significant relationship. Therefore, one in 20 of the significant results from data analysis could be significant by chance alone. Lowering the acceptable P value, or applying the Bonferroni correction, would lessen the chances of Type 1 errors and only reveal the strongest relationships, but it was considered to be more important to avoid Type 2 errors than Type 1 errors. In any case, the Bonferroni correction process is conceptually flawed, in that for any one relationship the chance of an incorrect verdict does not change whether it is alone in a sentence or grouped with a million other relationships in a mega-table.

As an exploratory process, the continuous variables that had significant correlations and the class variables that showed a significant relationship with the response variable were used in a 'best subsets' analysis in Minitab (Minitab Inc., 1998). A 'best subsets' analysis gives the two regression models with the highest level of explanation for each number of independent variables. The indicated models were tested using the general linear modelling procedure in Minitab Inc. (1998) which uses a regression approach by creating a: 'full rank' design matrix...from the factors and covariates...each response variable is regressed on the columns of the design matrix' (Minitab Inc., 1998, pp. 3–34). Continuous variables were used as covariates in the models, which used adjusted (Type III) sums of squares. A model with the highest adjusted R^2 was accepted if all components of the model were significant, and the residuals distributed, at least approximately, normally, as shown in a histogram. If this was not the case, the models with lesser levels of adjusted R^2 in the best subsets analysis were examined until one or none satisfied these requirements.

Due to the fact that most of the species recorded in gardens were insufficiently frequent to be satisfactorily modelled singly using the above procedure, a number of groups of species, based on origin, taxonomy and/or ecology, were formed: exotics; honeyeaters; natives; Psittaciformes; small native woodland birds (Table 1). The species that had their highest abundances in each classificatory group (Table 1) were regarded as an assemblage. The abundances of each assemblage were also used as dependent variables.

3. Results

Forty bird species were recorded utilising gardens, including 6 Tasmanian endemic and 6 exotic species. Garden bird species richness ranged from 0 to 18, with a mean of 5.2 ± 2.9 and a median of 5. Native bird species richness ranged from 0 to 16, with a mean of 2.9 ± 2.3 and a median of 2. Garden bird abundance ranged from 0 to 119, with a mean of 21.6 ± 18.0, and a median of 19. Native bird abundance ranged from 0 to 45, with a mean of 9.2 ± 8.8, and a median of 6. The exotic common blackbird (*Turdus merula*) was the most frequently recorded species, with 412 records. The exotic house sparrow

Table 1 – Mean abundances of bird species in garden types defined by birds

Species	Group				
	1	2	3	4	5
Noisy miner *Manorina melanocephala* (Latham)	**2.44**	0.66	0.00	0.14	0.00
Australian magpie *Gymnorhina tibicen* (Latham)	**0.33**	0.00	0.02	0.00	0.00
Musk lorikeet *Glossopsitta concinna* (Shaw) P	**0.33**	0.00	0.00	0.00	0.00
Galah *Cacatua roseicapilla* (Vieillot) P	**0.11**	0.00	0.00	0.00	0.00
Grey butcherbird *Cracticus torquatus* (Latham)	**0.07**	0.00	0.02	0.00	0.02
Laughing kookaburra *Dacelo novaeguineae* (Hermann)	**0.04**	0.00	0.00	0.00	0.00
Yellow-tailed black-cockatoo *Calyptorhynchus funereus* (Shaw) P	**0.04**	0.00	0.00	0.00	0.00
Little wattlebird *Anthochaera chrysoptera* (Latham)	0.11	**3.02**	1.73	0.43	0.70
Common starling *Sturnus vulgaris* (Linnaeus)[a]	1.67	**2.68**	1.70	0.50	0.60
Black-faced cuckoo-shrike *Coracina novaehollandiae* (Gmelin)	0.04	**0.07**	**0.07**	0.00	0.00
House sparrow *Passer domesticus* (Linnaeus)[a]	0.00	5.24	**18.92**	0.71	1.96
Silvereye *Zosterops lateralis* (Latham) w	0.00	1.83	**4.39**	0.11	3.57
Common blackbird *Turdus merula* (Linnaeus)[a]	2.33	3.66	**4.10**	2.14	2.85
European goldfinch *Carduelis carduelis* (Linnaeus)[a]	0.26	0.90	**1.05**	0.25	0.60
Black-headed honeyeater *Melithreptus affinis* (Lesson) w h	0.00	0.07	**0.86**	0.04	0.77
Eastern spinebill *Acanthorhynchus tenuirostris* (Latham) w	0.00	0.27	**0.46**	0.00	0.40
Superb fairy-wren *Malurus cyaneus* (Latham) w	0.00	0.00	**0.42**	0.00	0.36
European greenfinch *Carduelis chloris* (Linnaeus)[a]	0.00	0.32	**0.41**	0.11	0.15
Spotted turtle-dove *Streptopelia chinensis* (Scopoli)[a]	0.19	0.10	**0.37**	0.11	0.06
Eastern rosella *Platycercus eximius* (Shaw) P	0.07	0.00	**0.22**	0.07	0.09
Black currawong *Strepera fuliginosa* (Gould)	0.00	0.00	**0.12**	0.07	0.00
Forest raven *Corvus tasmanicus* (Mathews)		0.04	0.05	**0.07**	0.00
Grey shrike-thrush *Colluricincla harmonica* (Latham)	0.00	0.00	**0.02**	0.00	0.00
Scarlet robin *Petroica multicolor* (Gmelin) w	0.00	0.00	**0.02**	0.00	0.00
Welcome swallow Hirundo neoxena (Gould)	0.44	0.17	0.08	**0.57**	0.02
Masked lapwing *Vanellus miles* (Boddaert)		0.04	0.05	0.05	**0.14**
New Holland honeyeater *Phylidonyris novaehollandiae* (Latham) w h	0.07	0.10	2.86	0.00	**3.91**
Yellow wattlebird *Anthochaera paradoxa* (Daudin)	0.00	0.20	0.37	0.07	**0.92**
Brown thornbill *Acanthiza pusilla* (Shaw) w	0.00	0.05	0.31	0.00	**0.57**
Crescent honeyeater *Phylidonyris pyrrhoptera* (Latham) w h	0.00	0.29	0.42	0.04	**0.47**
Green rosella *Platycercus caledonicus* (Gmelin) P	0.00	0.00	0.20	0.00	**0.38**
Spotted pardalote *Pardalotus punctatus* (Shaw) w	0.00	0.22	0.14	0.21	**0.34**
Grey fantail *Rhipidura fuliginosa* (Sparrman) w	0.00	0.05	0.12	0.00	**0.21**
Striated pardalote *Pardalotus striatus* (Gmelin) w	0.04	0.00	0.05	0.07	**0.21**
Yellow-throated honeyeater *Lichenostomus flavicollis* (Vieillot) w h	0.00	0.00	0.14	0.00	**0.17**
Strong-billed honeyeater *Melithreptus validirostris* (Gould) w h	0.00	0.00	0.00	0.00	**0.11**
Swift parrot *Lathamus discolor* (Shaw) P	0.07	0.00	0.03	0.00	**0.11**
Grey currawong *Strepera versicolor* (Latham)	0.00	0.00	0.03	0.00	**0.09**
Golden whistler *Pachycephala pectoralis* (Latham) w	0.00	0.00	0.00	0.00	**0.06**
Tree Martin *Hirundo nigricans* (Vieillot)	0.04	0.05	0.03	0.00	**0.06**

The highest value for each species is shown in bold. Assemblages consist of the species that are shown in bold in the particular garden type. Group: 1 = edge assemblage; 2 = generalist assemblage; 3 = suburbanites assemblage; 4 = lawn assemblage; 5 = woodland assemblage. h = honeyeater; P = Psittaciforme; w = small woodland bird. Scientific and common names are shown.

a Exotic.

(*Passer domesticus*) was by far the most abundant, with 1458 individuals recorded. The most frequently recorded native bird was the New Holland honeyeater (*Phylidonyris novaehollandiae*), with 194 records. The most abundant native bird was the silvereye, with 534 individuals recorded. Other frequently recorded natives included the eastern spinebill (*Acanthorhynchus tenuirostris*) and the crescent honeyeater (*Phylidonyris pyrrhoptera*).

Gardens on the same property had less different assemblages of birds than all other pairs of gardens (ANOVA, $F = 71.26$, $df = 1$, $P < 0.001$). Garden characteristics were also less different between gardens on the same property than between all other pairs of gardens (ANOVA, $F = 195.34$, $df = 1$, $P < 0.001$). However, there was not a significant difference between the two sets of garden pairs in garden characteristic distance minus bird species composition distance (ANOVA, $F = 0.47$, $P = 0.495$). Therefore, treating back and front gardens on the same property as independent samples is considered to be justified.

Based on the ordination and classification analyses of bird species composition by garden, five groups of gardens were discriminated. One group of gardens was dominated by large, aggressive bird species characteristic of edges and open woodlands, the edge assemblage (Table 1). The galah (*Cacatua roseicapilla*), yellow-tailed black cockatoo (*Calyptorhynchus funereus*), musk lorikeet (*Glossopsitta concinna*), noisy miner (*Manorina melanocephala*), Australian magpie (*Gymnorhina tibicen*), grey butcherbird (*Cracticus torquatus*) and laughing kookaburra (*Dacelo novaeguineae*) all occurred most abundantly in this assemblage. The noisy miner, musk lorikeet and Australian magpie were particularly abundant. Several less aggressive and generally smaller birds, usually associated with woodland habitats, such as the silvereye, yellow wattlebird (*Anthochaera paradoxa*), eastern spinebill, black-headed honeyeater (*Melithreptus affinus*), superb fairy-wren (*Malurus cyaneus*) and brown thornbill (*Acanthiza pusilla*), were not present in group one whilst being relatively common in other groups.

A second group of gardens was characterised by relatively high abundances of medium sized birds that generally do not have highly specific habitat or feeding requirements, the generalist assemblage. Only two species, the common starling and little wattlebird (*Anthochaera chrysoptera*), were most abundant in this group. The black-faced cuckoo-shrike (*Coracina novaehollandiae*) was equally as abundant in groups 2 and 3. The noisy miner, silvereye, house sparrow and common blackbird were also abundant in group two.

Group 3 had high abundances of species that are well adapted to the garden environment, the suburbanites assemblage. Fourteen species were most abundant in group 3; this includes five out of the six introduced species recorded, house sparrow, European goldfinch (*Carduelis carduelis*), European greenfinch (*Carduelis chloris*), common blackbird and spotted turtle-dove (*Streptopelia chinensis*). The house sparrow was especially abundant in group 3. Other species with relatively high abundances in group 3 included the common blackbird, little wattlebird, New Holland honeyeater, silvereye and common starling.

Group 4 was best characterised by relatively high abundances of species that forage on lawns, the lawn assemblage. Only two species, the welcome swallow and masked lapwing (*Vanellus miles*), were most abundant in group 4. Other species, such as the common blackbird, common starling and little wattlebird were moderately abundant.

Group 5 was best characterised by the relatively high abundances of small native woodland species that are often displaced by urbanisation, the woodland assemblage. It was the most diverse of the five groups with 14 species occurring most abundantly. All 14 of these are native to Tasmania, including four endemic species, the green rosella (*Platycercus caledonicus*), yellow-throated honeyeater (*Lichenostomus flavicollis*), strong-billed honeyeater (*Melithreptus validirostris*) and the yellow wattlebird. Group 5 had relatively high abundances of several species that were generally uncommon in the other assemblages, such as the swift parrot (*Lathamus discolour*) which is a nationally endangered species, the grey fantail (*Rhipidura fuliginosa*), the strong-billed honeyeater and the striated pardalote (*Pardalotus striatus*).

The models for bird species richness and native bird species richness included garden area, canopy height and small shrub cover (Table 2). Richness was positively related to all these variables. The model for exotic bird species richness indicated a positive effect of supplementary feeding, large shrub cover and the floristic gradient in gardens that is related to rainfall. The total abundance of birds was best explained by high precipitation, supplementary feeding, presence of chicken yards, large garden area and large shrub cover.

The models for the five bird assemblages were highly individualistic, with altitude occurring in three, and annual precipitation, garden area and supplementary feeding occurring in two. Distance to Hobart, percentage cover of deciduous trees, chicken yard presence, dog presence, the native to exotic garden floristic gradient, percentage cover of small shrubs and canopy height all occurred in single models.

The models for the abundances of bird groups were again quite distinctive, with canopy height, percentage cover of small shrubs and garden area occurring in three models each, supplementary feeding and the native to exotic garden floristic gradient occurring in two models each and altitude, annual precipitation, chicken yard presence and the rainfall garden floristic gradient all occurring in single models.

Of the 12 individual species that were able to be modelled, five had annual precipitation in their models, four had supplementary feeding in their models, three had the native to exotic garden floristic gradient in their models, three had percentage cover of deciduous trees in their models, three had garden area in their models, three had distance to Hobart in their models, three had altitude in their models, two had distance to bush in their models, two had percentage cover of large shrubs in their models, two had canopy height in their models, two had percentage cover of small shrubs in their models, one had chicken yard presence in its model, one had permanent water in its model, one had dog presence in its model and one had percentage cover of evergreen trees (Table 2). Altitude, distance to Hobart, supplementary feeding and garden area were the most explanatory variable on two occasions each. Percentage cover of small shrubs was the most explanatory variable in one case. The rainfall garden floristic gradient, the percentage cover of lawn, the percentage cover of artificial surfacing, the percentage cover of

Table 2 – Components of models, showing tendencies and most significant variable

	Alt.	Dist. to Hobart	Annual precip.	Sup. feed.	Decid. tree cov.	Chick. yard	Gdn. area	Dog	Can. height	Sm. shrub cover	Lrg. shrub cover	Dist. to bush	Ever. tree cover	Perm. water	Fl.gra dient:nat. to exot.[1]	Fl.gradient: rainfall[2]	Lawn cover	Art. cover	Herb. cover	PSR
Garden bird species richness	1+	0	1+	1+	1+	0	X+[a]	0	X+	X+	1+	1–	1+	1+	0	1+	0	0	0	0
Native bird species richness	1+	1+	1+	0	1+	0	X+	0	X+[a]	X+	1+	1–	1+	1+	1+	0	0	1–	0	1
Exotic bird species richness	1 ∧	0	0	X+[a]	1+	0	1+	0	0	1+	X+	0	0	1+	0	X+	0	0	0	1
Garden bird abundance	1+	0	X+	X+[a]	1+	X+	X+	0	1+	1+	X+	0	1+	1+	0	1+	1+	0	0	1
Assemblage abundances (group number)																				
Edge assemblage (G1)	X–[a]	X+	1–	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Generalist assemblage (G2)	1–	0	X–[a]	X+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Suburbanites assemblage (G3)	1+	0	X+	X+[a]	X+	X+	X+	0	1+	1+	1+	0	1+	1+	0	1+	0	0	0	1
Lawn assemblage (G4)	X–[a]	1+	0	0	0	0	0	X+	0	0	0	1+	0	0	0	0	0	0	0	0
Woodland assemblage (G5)	X+[a]	0	1+	0	0	0	X+	0	X+	X+	1+	1–	1+	0	X+	0	1–	0	0	1
Group abundances																				
Exotics	0	0	0	X+[a]	1+	X+	1+	0	0	0	1+	0	1+	1+	0	X+	0	0	0	1
Honeyeaters	X+[a]	1–	1+	0	1+	0	X+	0	X+	X+	1+	1–	1+	0	1+	0	1–	0	0	1
Natives	1+	0	1+	X+	1+	0	X+	0	X+[a]	X+	1+	1–	1+	1+	X+	0	1–	0	0	1
Psittaciformes	0	0	0	0	0	0	1+	0	X+	1+	0	0	1+	0	X+[a]	0	0	0	0	1
Small woodland birds	1+	1–	X+	1+	0	0	X+[a]	0	1+	X+	1+	1–	1+	1+	1+	0	1–	0	0	1
Species' abundances																				
Black-headed Honeyeater	1+	0	X+[a]	X+	0	0	1+	0	0	0	0	1–	0	0	0	0	0	0	0	0
Common blackbird	1+	0	1+	1+	X+	0	X+[a]	0	1+	1+	1+	0	0	0	X–	0	0	0	1+	1
Eastern spinebill	0	0	1+	0	0	0	X+[a]	0	1+	0	0	X–	0	0	0	1+	0	0	0	1
European goldfinch	1∧	X+[a]	1∧	0	X+	0	0	0	0	0	X+	0	1–	1+	0	0	0	0	0	0
Green rosella	0	0	X+	0	0	0	0	0	X+	X+[a]	0	1–	0	0	0	0	0	0	0	1
House sparrow	1+	0	X+	X+[a]	1+	X+	1+	0	0	0	1+	0	1+	1+	0	0	0	0	0	1
New Holland honeyeater	X+[a]	1–	1+	0	0	0	0	0	X+	X+	X+	1–	1+	0	X+	0	1–	0	0	1
Noisy miner	X–[a]	X+	1–	0	1–	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Silvereye	0	1–	0	X+[a]	X+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Spotted turtle-Dove	0	X–[a]	0	X+	1+	0	0	0	0	0	0	0	0	X+	0	0	0	0	0	0
Superb fairy-wren	0	0	X+[a]	0	0	0	0	X+	1+	0	0	0	0	0	X+	0	1–	0	0	0
Yellow wattlebird	X+	1–	X+	0	0	0	X+	0	1+	0	0	X–	X+[a]	0	1+	0	1–	1–	0	1

1+ = tendency to favour native end of gradient, – = tendency to favour exotic end of gradient, 2+ = tendency to favour high rainfall end of gradient, – = tendency to favour low rainfall end of gradient. X = in model, O = not in model, 1 = not in model but statistically significant singly; + = tendency to increase with presence of or increase in variable, – = tendency to decrease with presence of or increase in variable, ∧ = no linear tendency (significance is difference between extremities and intermediate values), Alt. = Altitude. Dist. to Hobart = Distance to Hobart. Annual precip. = Annual precipitation. Sup. feed = Supplementary feeding. Decid. tree cov. = Deciduous tree cover. Chick. yard. = Chicken yard. Gdn. area = Garden area. Can. height = Canopy height. Sm. shrub cover = Percentage cover of small shrubs. Lrg. shrub cover = Percentage cover of large shrubs. Dist. to bush = Distance to bush. Ever. tree cover = Percentage cover of evergreen trees. Perm. water = Permanent water. Fl. gradients: nat. to exot. = Garden floristic gradient from native to exotic plants. Fl. gradient: low to hgh. = Garden floristic gradient from low rainfall to high rainfall. Lawn cover = Percentage cover of lawn. Art. cover = Percentage cover of artificial surfacing. Herb. cover = Percentage cover of herbaceous plants. PSR = Vascular plant species richness.
a Highest single relationship of model variables.

Table 3 – Number of incidences of native and exotic birds utilising local native, Australian native and exotic plant species in winter and spring

Birds/activity	Plants		
	Local native	Australian native	Exotic
Natives obtaining nectar	28	113	65
Natives eating fruit	2	2	17
Natives foliage gleaning	66	57	52
Natives bark gleaning	36	12	13
Natives nesting	0	2	4
Total native	132	186	151
Exotics obtaining nectar	0	2	0
Exotics eating fruit	0	0	30
Exotics foliage gleaning	1	1	2
Exotics bark gleaning	0	1	0
Exotics nesting	0	5	6
Total exotic	1	9	38

herbaceous plants, the presence of cats, and vascular plant species richness did not appear in any of the individual species models. In two models distance to bush had a negative relationship. Distance to Hobart and altitude displayed a negative relationship in one model each.

Native birds utilised local native plants, exotic Australian native plants and exotic non-Australian plants in almost equal incidence, while exotic birds utilised exotic non-Australian plants far more often than exotic Australian native plants and almost never utilised local native plants (Table 3).

4. Discussion

One of the major issues encountered when conducting a study on gardens is that participating landowners are generally those interested in gardening and/or conservation and thus the results are biased accordingly (Chamberlain et al., 2004; Zagorski et al., 2004). As participants in this project were not required to provide any information, or to participate in an interview, and because data collection was relatively unobtrusive, a very high percentage of the landowners who were approached were willing to participate. This enabled a wide range of garden types to be surveyed and makes the results of this study distinct from other similar studies that usually rely on data collected in the gardens of 'birders' (Moverly, 1997).

Altitude and rainfall had a substantial influence on bird species composition (Table 2). The fact that native species and assemblages were much more likely than exotic species and assemblages to have one or both of these environmental variables in their models may be partially a product of the substantial variation in the composition of the native plant component of gardens with climate and altitude (Kirkpatrick et al., in press). The exotic birds were dependent on introduced plants in gardens (Table 3). Gardens dominated by introduced plants, and fertilised and watered to support them, irrespective of environment, are ubiquitous in Hobart (Kirkpatrick et al., in press).

The two landscape variables had minimal influence on garden bird attributes (Table 2). When distance to natural vegetation was a significant influence, it was always a negative one, with bird abundance decreasing with increasing distance. Distance to Hobart had little effect on birds. The minimal influence of distance variables in Hobart is interesting as they have been widely found to influence urban bird communities (Green, 1986; Munyenyembe et al., 1989; Jokiamaki and Suhonen, 1993; Germaine et al., 1998; Fernandez-Juricic, 2000; Chamberlain et al., 2004), although Catterall et al. (1989) found that distance from native vegetation had very little influence on the suburban bird assemblages in Brisbane, Australia. The distance from city variable may have been relatively insignificant because garden size and style are less concentrically arranged in Hobart than in many other cities due to topographic constraints to urban growth. Similarly, natural vegetation is rarely more than a few km away from gardens.

The characteristics of the garden itself dominate most models, suggesting that the gardener can substantially influence bird species composition. Only the models for the edge assemblage and the model for the noisy miner lacked garden characteristics. The edge assemblage was very similar to the noisy miner-dominated assemblage that MacDonald and Kirkpatrick (2003) identified in remnant native vegetation in subhumid Tasmania. Noisy miners, and other edge species, aggressively exclude small native woodland birds (Grey et al., 1998; Ford et al., 2001; Major et al., 2001). The presence of such edge species had a dramatic affect on garden bird assemblages, with the absence of small woodland birds being marked. However, this was not the case for every garden in suburbs dominated by noisy miners. Small woodland birds, such as eastern spinebills, crescent honeyeaters and spotted pardalotes (*Pardalotus punctatus*), were recorded in small numbers in gardens with protective dense shrub layers. The dominance of noisy miners depends on the presence of relatively large areas of vegetation consisting of scattered trees with a short grassy understorey (MacDonald and Kirkpatrick, 2003). It is more a local landscape-scale effect than a garden-scale effect.

The positive relationship between bird species richness and area found in Hobart gardens has been recorded many times in natural and semi-natural vegetation (Ambuel and Temple, 1983; Freemark and Merriam, 1986; Bellamy et al., 1996; MacDonald and Kirkpatrick, 2003) and in gardens in Europe (Thompson et al., 1993).

The percentage cover of shrubs, both large and small, proved to be a very important influence on garden birds (Table 2). Shrubs favoured the small woodland birds, the woodland assemblage, and the New Holland honeyeater, reinforcing the notion that small native woodland birds are dependent on low vegetation cover for shelter in the urban environment (Green, 1986; White et al., 2005), as in natural vegetation (Wood, 1996; MacDonald and Kirkpatrick, 2003).

Gardens with tall trees were rich in native bird species and individuals, whereas gardens that had high covers of deciduous trees tended to favour many exotic birds (Table 2). Native birds showed a previously documented (Green, 1986; Green et al., 1989; Mills et al., 1989; Day, 1995; Parsons et al., 2006) preference for native plants, in that native birds tended to favour the more native end of the native-exotic floristic gradient (Table 2).

Supplementary feeding had a very strong positive relationship with exotic bird abundance and richness (Table 2). Supplementary feeding also increased native bird abundance, and the numbers of black-headed honeyeaters and silvereyes. The provision of seeds appeared to favour exotic species. A few garden owners supplied other types of supplementary food, such as bread, fruit and a nectar mix. These other food types, especially the nectar mix, appeared to favour the native birds. The strong influence of supplementary feeding reinforces the similar positive relationships found by Wilson (1994) and Chamberlain et al. (2004).

Permanent water was only significant enough to appear in one model, that for spotted turtle-dove abundance. Pet presence had little effect on the garden bird assemblages. Dog presence had a positive influence in the models of the lawn assemblage and the superb fairy-wren. Presumably, this is because of the fact that the superb fairy-wren, and one of the two birds most abundant in the lawn assemblage, forage predominantly on the ground, and are thus vulnerable to cat predation. Dogs tend to chase away cats, or at least spoil their predatory attempts, so their positive influence on ground foragers is logical.

As found in a study of Sydney gardens (Parsons et al., 2006), abundance of domestic cats had no significant relationship with any of the bird variables. This is not to say that cats do not prey on birds in Hobart. As elsewhere (Gillies and Clout, 2003), they do. Cats may not statistically influence garden bird characteristics because they are ubiquitous, with variation in their numbers related to variation in the abundance of their prey. The previously undocumented positive effect of chicken yards on many garden bird characteristics would not be perceptible if all gardens had fowl.

Whilst native birds tended to be generally more abundant in gardens dominated by Australian native plants, they were also abundant in some gardens with few such plants. Interestingly, it was not just the adaptable nectar and fruit feeders among the native birds that used exotic plants as resources, but also some birds that were generally uncommon, such as those in the woodland assemblage and small woodland birds. Thus, the popular hypothesis that the lack of native plants in suburban areas is the proximal cause of the rarity of small woodland birds is questionable. Many non-local Australian natives, in particular, species in the genera *Grevillea*, *Eucalyptus* and *Acacia* were very well used by native birds, as were non-Australian exotic taxa, such as *Betula pendula*, *Prunus* spp., *Echium candicans* and *Cotoneaster* spp. It would clearly be possible to have close to a full assemblage of small native woodland birds in a garden comprised totally of non-local native plant species. In contrast, the exotic birds almost totally utilized non-Australian plant species.

5. Conclusions

The floristic composition, structure and other attributes of Hobart gardens can be manipulated to favour particular sets of bird species. Local native gardens are desirable, but not necessary, to gain a healthy complement of local native birds. These findings, and the fact that the woodland bird assemblage, in particular, contained endemic, threatened and uncommon native bird species, show that domestic suburban gardens can be important in the conservation of urban avifauna. Further research is needed to reveal if gardens in other places can be equally as important as Hobart gardens in the conservation of urban native avifauna.

The analyses suggest that the most native bird-friendly garden would be large, have many tall native evergreen trees and a dense native shrub understorey. Exotic birds could be discouraged by the avoidance of: supplementary feeding with seeds; chicken yards; and exotic, especially deciduous, plants. However, it needs to be emphasized that, if all gardens were native-friendly and exotic-hostile as described above, there would be many native bird species that would find the suburbs of Hobart a less friendly environment than at present, including the raptors that prey on the exotic birds and animals attracted by chicken yards. At the suburb scale, diversity in garden types is likely to maximise native bird species richness, and, at the garden scale, an abundance of some exotic birds may be a necessary consequence of promoting the interests of some native species.

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